

# Data Center Cooling Market 2026-2035 | Size, Growth, Industry Trends and Report

The global data center cooling market has emerged as a critical segment within digital infrastructure development, driven by the exponential growth of data generation and cloud computing. In 2025, the market was valued at USD 26.7 billion and is projected to expand significantly, reaching USD 126.3 billion by the end of 2035. This expansion reflects a strong compound annual growth rate (CAGR) of 16.8% during the forecast period from 2026 to 2035.

The rapid scale-up of hyperscale data centers, increasing adoption of artificial intelligence workloads, and rising energy density within server racks have elevated cooling from a supporting function to a strategic investment area. As data centers become more complex and power-intensive, advanced cooling solutions are gaining prominence as essential enablers of performance, sustainability, and operational continuity.

## Data Center Cooling Industry Demand

### Market Description

[Data center cooling](#) refers to a range of technologies, systems, and services designed to regulate temperature, humidity, and airflow within data center environments. These solutions ensure optimal operating conditions for servers, storage systems, and networking equipment, preventing overheating, system failures, and unplanned downtime.

Modern data center cooling solutions have evolved beyond traditional air conditioning to include liquid-based, evaporative, and hybrid cooling technologies. These innovations are increasingly tailored to different data center architectures, including enterprise, hyperscale, colocation, edge, and modular facilities.

### Industry Demand Drivers

Demand for data center cooling solutions is driven by the need for cost-effective thermal management, as cooling accounts for a substantial portion of data center operational expenses. Energy-efficient cooling technologies help operators reduce electricity consumption while maintaining high system availability.

Ease of administration is another key factor, as automated cooling systems equipped with intelligent monitoring and control capabilities reduce manual intervention and maintenance complexity. Additionally, data center cooling infrastructure offers a long operational lifespan, making it a durable and scalable investment for long-term digital infrastructure planning.

The surge in cloud services, remote work, streaming platforms, and AI-driven applications has significantly increased server density, intensifying the need for reliable and efficient cooling solutions across global data center networks.

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## **Data Center Cooling Market: Growth Drivers & Key Restraint**

### **Growth Drivers**

- **Rapid Expansion of Hyperscale and Cloud Data Centers**  
The accelerating shift toward cloud computing and digital services has led to the construction of large-scale hyperscale data centers. These facilities require advanced cooling solutions capable of managing high heat loads while maintaining energy efficiency, driving sustained demand for innovative cooling technologies.
- **Technological Advancements in Cooling Systems**  
Advancements such as liquid immersion cooling, direct-to-chip cooling, AI-driven thermal optimization, and modular cooling architectures are transforming the market. These technologies enable higher rack densities, improved energy efficiency, and better scalability, supporting next-generation computing workloads.
- **Rising Focus on Energy Efficiency and Sustainability**  
Data center operators are under increasing pressure to reduce carbon emissions and meet sustainability targets. Energy-efficient cooling solutions, including free cooling and evaporative systems, play a crucial role in minimizing environmental impact and operational costs.

### **Restraint**

- Despite strong growth momentum, high initial capital investment remains a major restraint. Advanced cooling technologies often require significant upfront costs and specialized infrastructure, which can be a barrier for small and mid-sized data center operators.

## **Data Center Cooling Market: Segment Analysis**

### **Segment Analysis by Component**

Solutions form the core of the data center cooling market, encompassing hardware and software systems designed to regulate thermal conditions. These solutions are widely adopted due to their direct impact on performance and energy efficiency.

Services, including installation, maintenance, consulting, and monitoring, support the long-term operation and optimization of cooling systems and are gaining traction as data centers seek outsourced expertise.

### **Segment Analysis by Type of Cooling**

Air-based cooling remains widely used across enterprise, colocation, edge, and modular data centers due to its familiarity and ease of deployment.

Liquid-based cooling is gaining rapid acceptance, particularly in hyperscale, colocation, and modular data centers, as it efficiently handles high-density computing workloads.

Evaporative cooling is increasingly adopted in regions with favorable climates, offering energy-efficient temperature control.

Other cooling types, including hybrid and emerging technologies, address specialized operational requirements across diverse data center environments.

### **Segment Analysis by Product**

Precision Air Conditioning (PAC) units continue to be a foundational cooling solution, especially in traditional enterprise data centers.

Chillers and cooling towers support large-scale facilities by managing heat rejection efficiently.

Economizers are gaining popularity due to their ability to reduce energy consumption through free cooling methods.

Liquid cooling systems are experiencing strong growth as demand rises for high-performance computing and AI applications. Other products contribute to customized and niche cooling requirements.

### **Segment Analysis by Data Center Type**

Enterprise data centers emphasize reliability and control, driving steady demand for traditional and hybrid cooling systems.

Colocation data centers prioritize scalability and energy efficiency, supporting the adoption of modular and liquid cooling solutions.

Hyperscale data centers represent a key growth segment, demanding highly efficient, large-scale cooling architectures.

Edge and modular data centers require compact, flexible, and rapidly deployable cooling solutions, influencing product innovation.

### **Segment Analysis by End User Industry**

The IT & Telecom sector dominates demand due to continuous expansion of digital services and network infrastructure.

BFSI relies on robust cooling to support data security and high-availability systems.

Healthcare data centers support digital health records and imaging systems, driving demand for reliable cooling.

Energy, retail, and other industries are increasingly adopting data centers to support analytics and automation initiatives.

## **Data Center Cooling Market: Regional Insights**

### **North America**

North America represents a mature and technologically advanced market for data center cooling. Strong cloud adoption, widespread hyperscale data center deployment, and stringent energy efficiency standards drive demand. Continuous innovation and investment in sustainable cooling solutions support market expansion.

## **Europe**

Europe's market is shaped by regulatory emphasis on energy efficiency and environmental sustainability. Rising investments in green data centers and increasing digital transformation across industries drive demand for advanced cooling technologies tailored to regional climate and regulatory requirements.

## **Asia-Pacific (APAC)**

APAC is one of the fastest-growing regions, driven by rapid digitalization, expanding internet penetration, and large-scale data center construction. Growth in cloud services, e-commerce, and smart city initiatives fuels strong demand for scalable and cost-efficient cooling solutions.

## **Top Players in the Data Center Cooling Market**

The data center cooling market is highly competitive, with leading players focusing on innovation, sustainability, and global expansion. Key companies operating in the market include Vertiv (U.S.), Schneider Electric (France), STULZ (Germany), Johnson Controls (U.S.), Munters (Sweden), Trane Technologies (U.S.), Carrier Global Corporation (U.S.), Rittal (Germany), Mitsubishi Electric (Japan), Fujitsu (Japan), Hitachi (Japan), Nortek Air Solutions (U.S.), Airedale International Air Conditioning (UK), CoolIT Systems (Canada), Green Revolution Cooling (GRC) (U.S.), LiquidStack (U.S.), Alfa Laval (Sweden), Samsung Electronics (South Korea), Submer (Spain), and Delta Electronics (Taiwan).

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